An Architecture for a Network Anomaly Detection Framework

draft-ietf-nmop-network-anomaly-architecture-07
draft-ietf-nmop-network-anomaly-semantics-05
draft-ietf-nmop-network-anomaly-lifecycle-05

Motivation and architecture of a Network Anomaly Detection Framework
and the relationships to other documents describing
network symptom semantics and network incident lifecycle

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Problem Statement and Motivation

How it is being addressed in each document

Network Anomaly Detection

To detect service interruptions before users do, network operations must use **automated**, **holistic** monitoring across all **three network planes**.

- draft-ietf-nmop-network-anomaly-architecture describes the motivation, architecture and the relationship to other two documents.

By automating the postmortem process and standardizing labeled incident data, operators can **improve machine learning** models continuously and **foster better collaboration** between vendors and academia.

- draft-ietf-nmop-network-anomaly-semantics defines Symptom semantics to enable standardized data exchange to validate results with network engineers and improve supervised and semi-supervised machine learning systems.
- draft-ietf-nmop-network-anomaly-lifecycle describes the lifecycle process, enabling network engineers to interact with the network anomaly detection system to refine the detection abilities over time.

Network Anomaly Detection Architecture

DRAFT-IETF-NMOP-NETWORK-ANOMALY-ARCHITECTURE

Network Anomaly Detection Architecture

Document summary

As human-driven troubleshooting becomes less **scalable** due to its vast number of dependencies, this draft outlines a **standardized**, **architecture-level framework** designed to **automate the detection of service disruptions.**

```
    (1)-------+                      (11)----------------+
    | Service |                      |     Alarm and     |
|-- |Inventory|                      | Problem Management|
|   |         |                      |      System       |
|   +---------+                      +-------------------+
|     |                                     ^     Stream
|     |                                     |
|     |       (12)------+            +-------------------+
|     |       | Post-   | Stream     |   Message Broker  |
|     |       | mortem  | <--------  |  with Analytical  |
|     |       | System  |            |    Network Data   |
|     |       +---------+            +-------------------+
|     |            |                        ^     Stream
|     |            |                        |
|     | (8)        | (3)            +-------------------+ Store
|     | Profile    | Fine           | Alarm Aggregation | Label
|     | and        | Tune           | for Anomaly       | --------|
|     | Generate   | SDD            | Detection         |         |
|     | SDD Config | Config         +-------------------+         |
|     |            |                       ^  ^  ^ Stream         |
|     v            v                       |  |  |       Replay   v
|  (2)-----------------+ (9)        (6)-----------------+    (10)------+
|  | Service Disruption| Schedule   | Service Disruption|    |  Data   |
|  |     Detection     | ---------> |     Detection     |<---| Storage |
|  |   Configuration   | Strategy   |                   |    |         |
|  +-------------------+            +-------------------+    +---------+
|                                      ^ ^ Stream ^ ^ ^           ^
|                                      | |        | | |           |
|                                   (7)-------(5)-------+         |
|                                   | Network |  Data   | Store   |
|--------------------------------> |  Model  |  Aggr.  | --------|
                                    |         | Process | Operational
                                    +---------+---------+ Data
                                           ^  ^  ^ Stream
                                           |  |  |
                                    +-------------------+
                                    |   Message Broker  |
                                    |  with Operational |
                                    |    Network Data   |
                                    +-------------------+
                                           ^  ^  ^ Stream
Subscribe                   Publish        |  |  |
      +-------------------+         (4)-----------------+
      | Network Node with | ------> | Network Telemetry |
----> | Network Telemetry | ------> |  Data Collection  |
      |   Subscription    | ------> |                   |
      +-------------------+         +-------------------+
```

Figure 1: Service Disruption Detection System Architecture

draft-ietf-nmop-network-anomaly-architecture

Network Anomaly Detection Architecture

Key draft updates since v-05

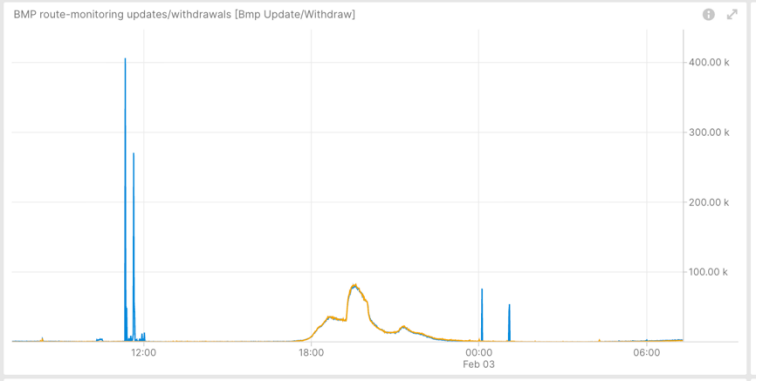
- **Terminology update**
  - Change from “customer profile” to “service profile” to ensure consistency across reference documentation (RFC 8969).
  - Added formal “Rules" definition to distinguish between knowledge-based approach and ML approach.
- **Restructure**
  - Reorganisation Service Disruption Section to improve reading
- **Refinement**
  - Clarified the relationship between Semantic, Rule, & Knowledge-based detection.
  - Refined definition: collective outlier, confidence score, concern score.
  - Refined clarification: Alarm-raising criteria, Data storage considerations, Operational data aggregation
- Explicitly **out-scoped** Service Degradation based on WG review feedback

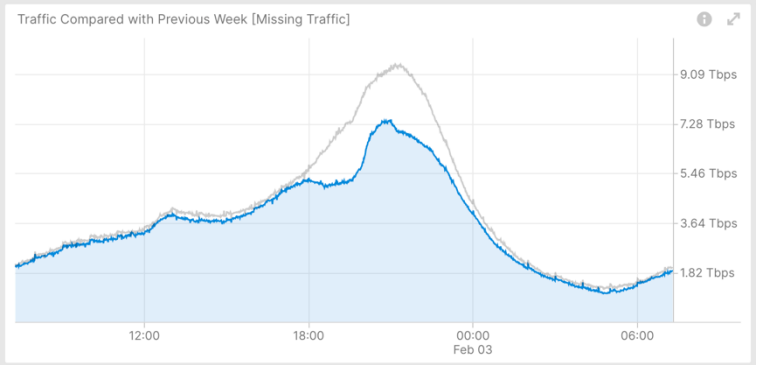
Network Anomaly Detection Architecture

Document implementation and operational experience

L3 VPN – Real-Time Incident Analysis

Swisscom TV service was severely impaired during 6 hours due to hardware faults and a software defect, which together led to a DoS condition.



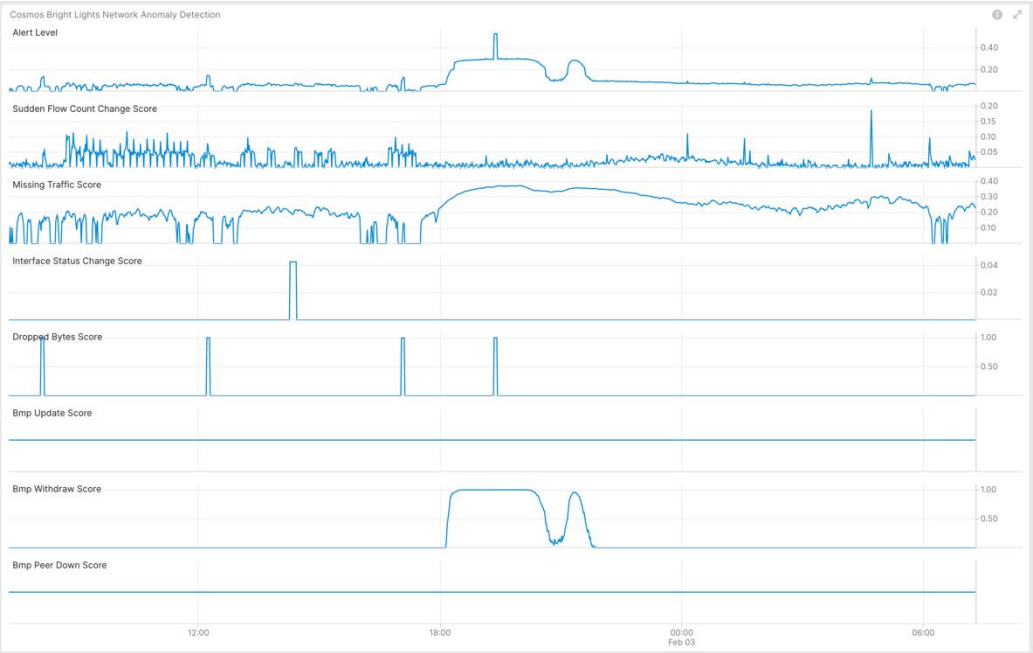


Network Anomaly Detection Architecture

Document implementation and operational experience

L3 VPN – Real-Time Incident Analysis

Concern Score: 0.53
Flow Count Spike: 0.03 Missing Traffic: **0.37** Traffic Drop: **1.00**
BMP Peer: 0.00 Interface Down: 0.00
BMP Update: 0.00 BMP Withdrawal: **1.00**



- **BMP route-monitoring Update/Withdraw check detected continuous and excessive routing topology changes due to users restarting tv boxes.**
- BMP peer Down/Up check did not apply because the hardware crash happened at the application layer.
- Interface Down/Up check did not apply.
- **Traffic Drop spike recognized drops due to unstable routing topology.**
- **Missing Traffic** detected.
- Sudden increased or decreased Flow Count was not applicable.

Overall: data plane and control plane checks have detected the excessive routing topology changes with service disruption.

Semantic Metadata Annotation

DRAFT-IETF-NMOP-NETWORK-ANOMALY-SEMANTICS

Semantic Metadata Annotation

Document summary

To facilitate the **exchange** of network anomaly data across diverse systems, and **benchmarking** of various anomaly detection systems, this draft defines a **standardized semantic metadata schema** and a set of **semantically structured terms** that provide human-readable context for network anomalies.

Action	Reason	Trigger
Interface State	Up	Link-Layer
Interface State	Down	Link-Layer
Interface Statistics	Errors	-
Interface Statistics	Discards	-
Interface Statistics	Unknown Protocol	-

Table 3: Description of symptoms and their actions, reasons and triggers for Management Plane.

Semantic Metadata Annotation

Key draft updates since v-03

- **Terminology update**
  - Added missing definition of terminologies used in the draft such as concern score and symptom.
- **Schema change**
  - Removed unused typedef score
  - Refined vpn-node-termination
  - Introduced dedicated L2 & L3 groupings for service
- **Symptom table update**
  - Update symptom: reason “Previous” -> “Drop”, trigger “Time” -> “Outside Monitored Domain”
- Avro schema changed accordingly

```
augment /rsn:relevant-state/rsn:anomaly:
      +--rw vpn-node-terminations* [hostname vrf-name]
         +--rw hostname                inet:host
         +--rw vrf-id?                 uint32
         +--rw vrf-name                string
         +--rw route-distinguisher?    string
         +--rw interface-id*           uint32
         +--rw interface-name*         string
         +--rw peer-ip*                inet:ip-address
         +--rw next-hop*               inet:ip-address
```

```
augment /rsn:relevant-state-notification/rsn:service:
      +--:(l2vpn)
      |  +-- l2vpn-service* [vpn-id]
      |     +-- vpn-id                string
      |     +-- uri?                  inet:uri
      |     +-- vpn-name?             string
      |     +-- site-ids*             string
      |     +-- change-id?            yang:uuid
      |     +-- change-start-time?    yang:date-and-time
      |     +-- change-end-time?      yang:date-and-time
      +--:(l3vpn)
         +-- l3vpn-service* [vpn-id]
            +-- vpn-id                string
            +-- uri?                  inet:uri
            +-- vpn-name?             string
            +-- site-ids*             string
            +-- change-id?            yang:uuid
            +-- change-start-time?    yang:date-and-time
            +-- change-end-time?      yang:date-and-time
```

Semantic Metadata Annotation

Document implementation and operational experience

Snippet of message example from Cosmos Bright Lights Implementation


```
 "vpnNodeTerminations": [
    {
        "hostname": "pcb05ro1010bew",
        "vrfId": null,
         " vrfName": "null",
        "routeDistinguisher": {
        "string": "0:64499:100196518"
    },
    "peerIp": [
        "138.187.124.33",
        "138.187.124.35"
    ],
    "nextHop": [],
    "interfaceId": [],
    "interfaceName": []
   }
 ],
 "stage": "detection",
 "operationalData":
  {
     " topicName”: null,
     " subjectName”: null
  }
```

```
"service": {
    "L3VpnServiceContainer": {
        "l3VpnService": [
            {
                "vpnId": "64497:59121",
                "uri": {
                    "string": "https://thor-
ui.thoruipp.corproot.net/cantata/lcs?dstCommunity=64497:59121"
                },
                "vpnName": {
                    "string": "64497:59121"
                },
                "siteIds": null,
                "changeId": null,
                "changeStartTime": null,
                "changeEndTime": null
            }
        ]
    }
  }
```

Network Anomaly Lifecycle

DRAFT-IETF-NMOP-NETWORK-ANOMALY-LIFECYCLE

Network Anomaly Lifecycle

Document summary

To bridge the gap between **automated network anomaly detection** and **actionable operational tasks**, this draft defines a standardized, **iterative lifecycle** designed to improve the **correctness** of anomaly detection system through **human-in-the-loop feedback**.

```
                          +-------------+
              +--------> |  Detection  | ---------+
 Adjustments |           +-------------+          | Symptoms
             |                                    |
             |                                    v
     +------------+                        +------------+
     | Refinement |<---------------------- | Validation |
     +------------+        Problem         +------------+
                         Confirmation
```

Figure 1: Anomaly Detection Refinement Lifecycle

```
       +-------+                     +---------+
       | Final |                     | Initial |----------------+
       +---^---+                     +---------+                |
           |                             |                      |
           |                         +-----+---------+          |
+-----------------------+     +--------|---------------|------+   |
|  +-----------------+  |     | +------v-----+  +------v----+ |   |
|  |    Detection    |-------->| |  Problem   |  |  Problem  | |   |
|  |     Adjusted    |  |     | | Forecasted |  | Potential | |   |
|  +--------^--------+  |     | +------------+  +-----------+ |   |
|           |           |     +--------|--Detection---|-------+   |
|           |           |             |              |           |
|           |           |             +------- ----- +           |
|           |           |                     |                  |
|           |           |                     |                  |
|           |           |                     |                  |
|           |           |                     v                  |
| +---------|--------+  |     +-----------Validation------------+ |
| |     Problem      |  |     |  +-----------+                 | |
| |     Analyzed     |<-------|  |  Problem  |   |  Problem  | | |
| +------------------+  |     |  | Discarded |   | Confirmed |<-|--+
|                       |     |  +-----|-----+   +-----------+ |
+-------Refinement------+     +-------------------------------+
                                      |
                                  +---v---+
                                  | Final |
                                  +-------+
```

Figure 2: Network Anomaly State Machine

Network Anomaly Lifecycle

Key draft updates since v-03

- **Terminology update**
  - Renamed “state/phase” to “stage” to avoid confusion with protocol “states”.
- **Schema change**
  - Changed “state” -> “stage”
  - Fixed RFC 8407 related Yang warnings (remove mandatory false)
  - Include operational data in anomaly and introduce “message-broker-grouping”
- **Refinement**
  - Refined definition of confidence score and concern score
  - Optimized Figure 2
  - Consolidated the usage of score across different drafts

```
 leaf stage {
         type identityref {
           base network-anomaly-stage;
         }
         mandatory true;
         description
           "Stage of the relevant state.";
       }
```

```
grouping message-broker-grouping {
       description
         "References message broker draft-ietf-nmop-yang-
message-broker-integration entities.";
       leaf topic-name {
         type inet:host-name;
         description
           "From which message broker topic the
operational data was consumed from.";
       }
       leaf subject-name {
         type inet:host-name;
         description
           "To which message broker subject the
operational data references to.";
       }
     }
```

Next Steps and Remaining Issues

Feedback on latest changes

Next Steps

- Requesting working group feedback on the editorial changes.
- Request YANG doctors review for draft-ietf-nmop-network-anomaly-semantics-05 and draft-ietf-nmop-network-anomaly-lifecycle-05.
- Implement semantic metadata in Cosmos Bright Lights to replace placeholder values (null) for newly introduced fields, including site-ids, interface names, and VRF names.
- Extend the postmortem system requirements, accompanied by an implementation proof of concept

Remaining Issue

- No known remaining issues at this time.